Jose Emilio Atauchi Mamani

Informatics and Systems Engineer · Cusco, Peru

ataauchimamani@gmail.com | +51 929618079 | LinkedIn | FiboNacci-codder

Codeforces: FiboEm | ICPC profile | Lattes: 8640962660930319

PROFILE

Informatics and Systems Engineering graduate from UNSAAC (2025) working on information retrieval and natural language processing. I am currently building a local Retrieval-Augmented Generation (RAG) pipeline for institutional documents at an electric utility, where I evaluate OCR, embedding and large language models. My background comes from algorithms and competitive programming (ICPC Latin America Regional, 2022–2024).

RESEARCH INTERESTS

Large Language Models · Retrieval-Augmented Generation · Information Retrieval · Natural Language Processing · Algorithms

EDUCATION

B.S. in Informatics and Systems Engineering 2020 – 2025

Universidad Nacional de San Antonio Abad del Cusco (UNSAAC), Cusco, Peru

- Thesis work linked to a RAG system for institutional documents.

Relational Database Certification Jan. 2026

freeCodeCamp

RESEARCH & PROFESSIONAL EXPERIENCE

ICT Professional Intern Feb. 2026 – Present

Electro Sur Este S.A.A. (ELSE), Cusco, Peru

- Designed and evaluated a RAG system that answers natural-language questions over 72 internal documents of a Peruvian electricity distributor, running entirely on **open-weight models**.
- Ingestion: structured OCR with **TeleOCR**, cleaning by block type, hierarchical-section chunking and indexing of dense and sparse vectors in **Qdrant**.
- Retrieval: query spelling correction and rewriting (Qwen3.5-4B), **hybrid search** (harrier-oss-v1-0.6b embeddings + BM25) fused with **RRF**, and cross-encoder reranking (Qwen3-Reranker-8B).
- Built a golden set of 170 manually verified questions and evaluated retrieval (Acc@k, Recall@10, MRR, nDCG) and hallucination with an LLM judge (gpt-oss-120b).
- Compared 9 chunking strategies (hierarchical section reached the best Recall@10, 0.959) and 10 generator LLMs; selected **Gemma 4 26B-A4B** (15.3% hallucination, 97.6% correct rejection of unanswerable questions, 0.82 s).
- Currently working on **tool calling** for the LLM.

IT Intern 2024 – 2025

Red de Comunicaciones UNSAAC (RCU), Cusco, Peru

- Web development, network administration and IT management support (30 h/week).
- Configuration and maintenance of routers in the university network; hardware and software technical support.

RESEARCH & DEVELOPMENT PROJECTS

- **Facial Recognition Model for Anxiety Indicators** 2024
  Computer vision model to identify facial patterns and expressions associated with high anxiety levels, aimed at early alerts for preventive care. (Coordinator)
- **AstroBioHub – NASA Space Apps Challenge** 2025
  AI knowledge engine for NASA space biology publications using NLP, semantic search and knowledge graphs. Role: DevOps engineer (deployment and infrastructure).
- **Chancay Megaport and International Tourism in Cusco** 2024
  Time-series forecasting with **LSTM** and **ARIMA** on MINCETUR data to analyze the megaport's impact on international tourist flow to Cusco.
- **Gaussian Elimination in Assembly vs. High-Level Language** 2024
  Execution-time comparison of Gaussian elimination implemented in assembly with **AVX** instructions and in Python.
- **Quantum Decryption and RSA Algorithm Analysis** 2024
  Study of RSA and classical encryption and their vulnerability to quantum decryption techniques.
- **Machine Learning for Native Potato Varieties in Cusco** 2023 – 2024
  ML models to recognize and classify native potato varieties of the Cusco region.
- **Knapsack Problem in CUDA** 2022 – 2023
  Parallel GPU implementation (**CUDA**) of the knapsack problem, an NP-complete problem, showing performance gains over sequential approaches.

- **Automated Attendance System** 2024
  Face-recognition attendance with FaceNet, Keras and OpenCV. [GitHub]

COMPETITIONS & AWARDS

- **NASA Space Apps Challenge** – 2nd place (Cusco) and Global Nominee 2025
- **ICPC Latin American Regional Contest** – 42nd in Latin America (2024), 62nd (2023) 2023 – 2024
- **ICPC South America/South Finals** – Participant 2022
- **Training Camp Argentina** (Rosario) – Participant 2024
- **IEEEXtreme** 18.0 and 19.0 – Participant 2024 – 2025
- **Ibero-American University Mathematics Olympiad (OIMU)** – Participant 2025
- **SUNASS Datathon 2026** – Southern Macro-region, data analysis for SEDACUSCO 2026
- **Cuscontest** (UNSAAC programming contest) – editions XIX, XX, XXI, XXV 2023 – 2026
- **Peruvian National Mathematics Olympiad (ONEM)** – Finalist (national round), representing Cusco 2019

TECHNICAL SKILLS

- **AI / IR:** RAG, LLMs, tool calling, embeddings, hybrid search (BM25, RRF), reranking, OCR, document processing, NLP, model evaluation.
- **Libraries & tools:** TeleOCR, Docling, LangChain, Qdrant, vLLM, RAGAS, Hugging Face, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Django.
- **Programming:** Python, C, C++, CUDA, Bash, SQL, HTML/CSS.
- **Databases:** PostgreSQL, SQL Server, Qdrant.
- **Infrastructure:** Linux, Git, Docker, Apptainer, Conda, NVIDIA CUDA.

ACTIVITIES

- Member of the **Association for Computing Machinery (ACM)** since 2022.
- Active member of the **ACM Cusco Study Circle** and the **OmegaUp Cusco Club**.
- Attendee, **Google Student Summit Peru** (2024).

LANGUAGES

Spanish: Native **English:** Intermediate **Portuguese:** Intermediate